

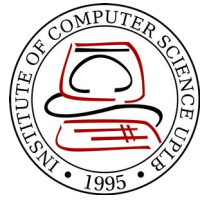
CMSC 137

Data Communications and Networking

Lecture Module
15 – Media Access Control



University of the Philippines
LOS BAÑOS



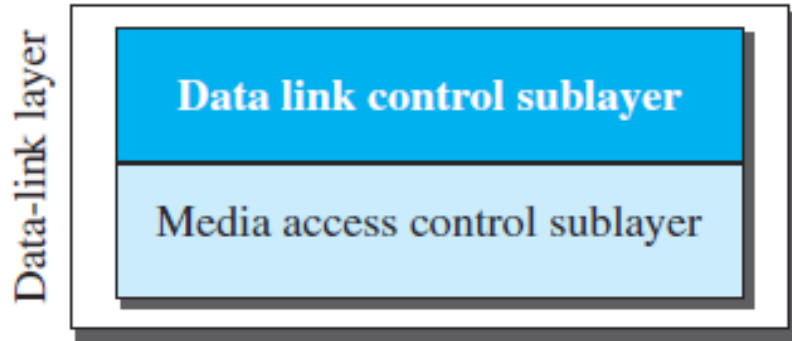
Module 15: Media Access Control

Rozano S. Maniaol
Lecturer

Media Access Control (MAC)



- Media Access Control – responsible for resolving access to the shared link

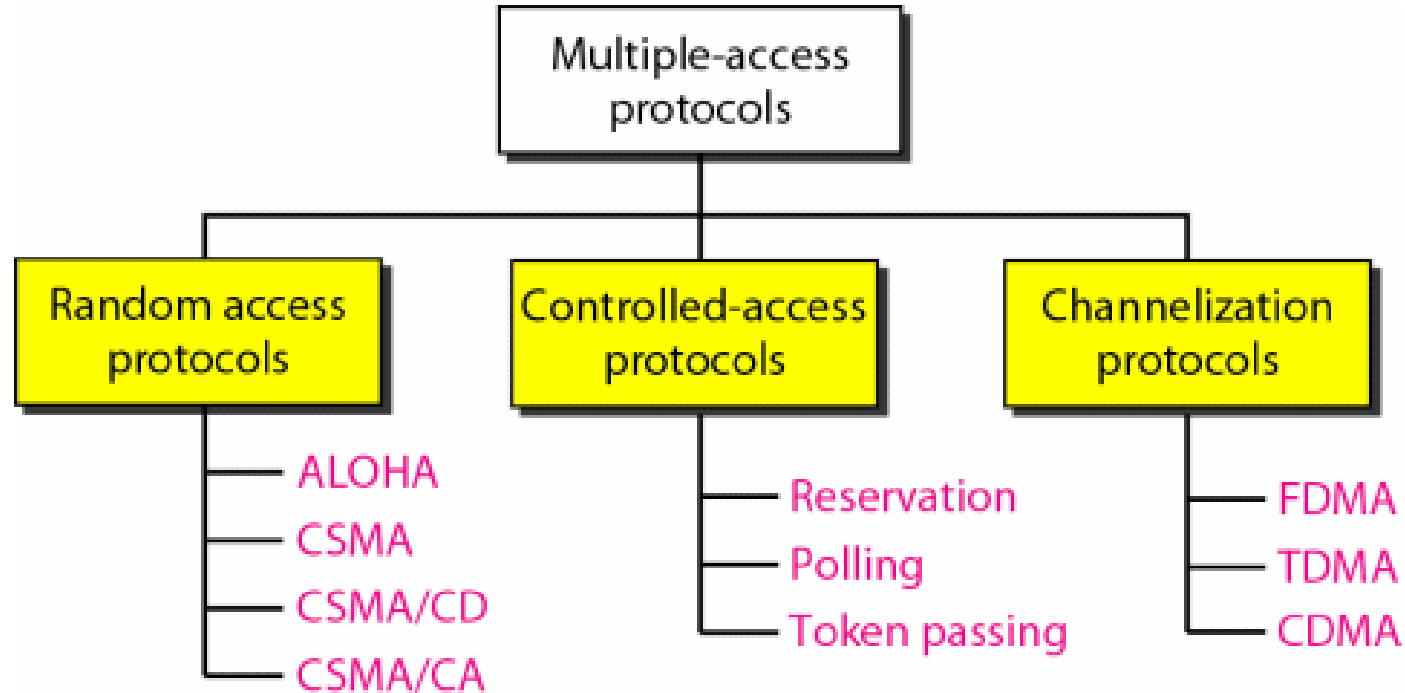
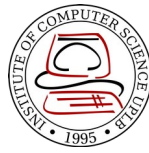


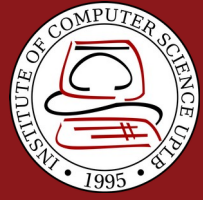
a. Data-link layer of a broadcast link



b. Data-link layer of a point-to-point link

Multiple Access Protocols





Random Access Protocols

Random Access



- Aka contention methods
- No station is superior to another station and none is assigned the control over another
- Each station that has data to send uses a procedure defined by the protocol to make a decision on whether or not to send
- Decision depends on the state of the medium
- No scheduled time for a station to transmit
- No rules specify which station should send next

Random Access



- If more than one station tries to send, there is an access conflict, called collision
 - When can the station access the medium?
 - What can the station do if the medium is busy?
 - How can the station determine the success or failure of the transmission?
 - What can the station do if there is an access conflict?

ALOHA

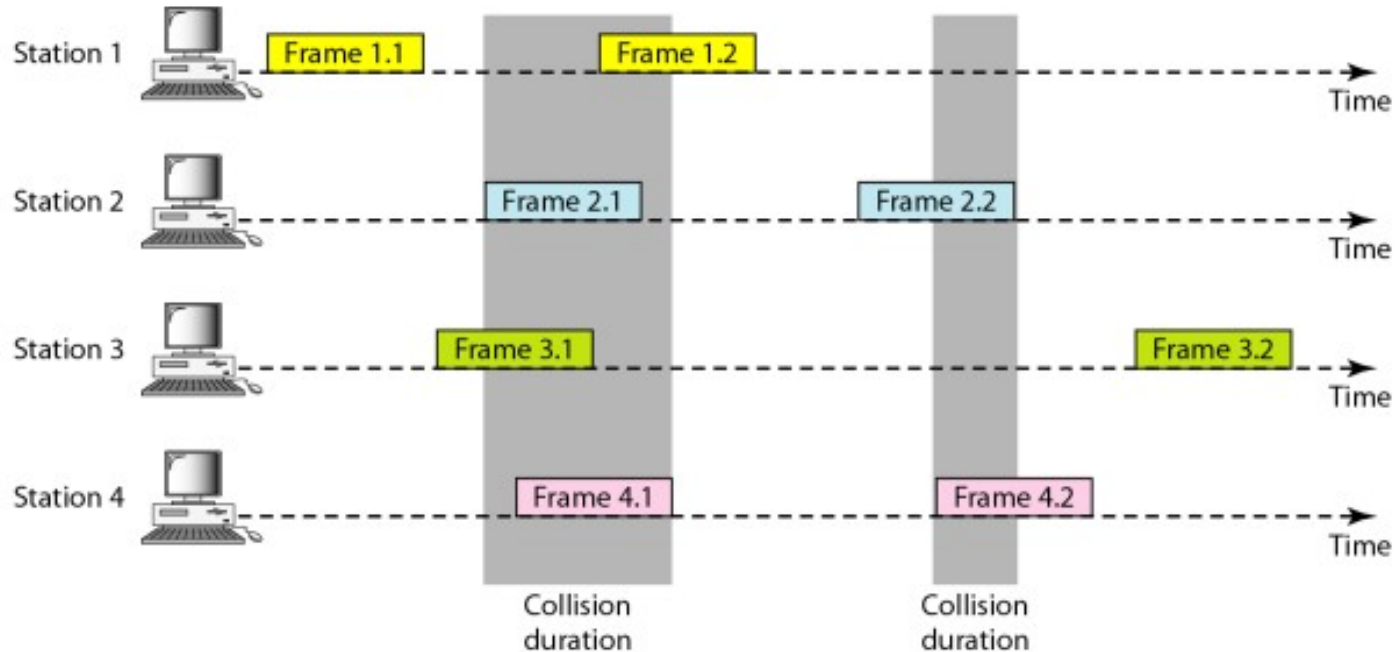


- Earliest random access method
- Developed at the University of Hawaii, 1970's
- Designed for radio LAN
- Medium (wireless) is shared between stations, collisions may happen

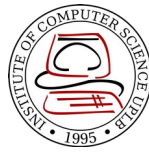
Pure ALOHA



Send whenever there is a frame to send



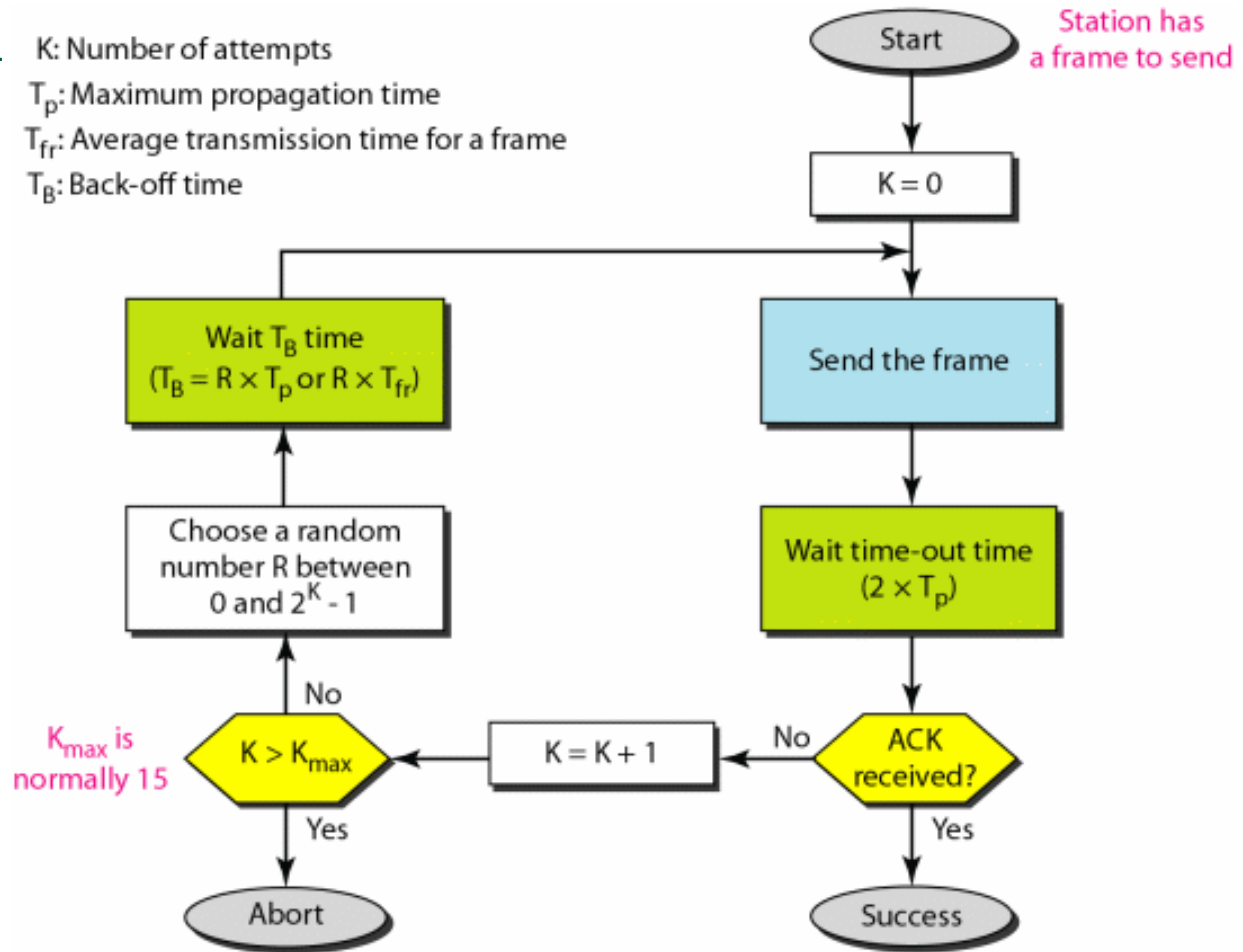
Pure ALOHA



- Relies on ACKs
- When the time-out period passes, each stations waits a random amount of time before re-sending its frame, **back-off time**, T_B
- After a maximum number of re-transmission attempts K_{max} , a station must give up and try later
- **Binary exponential back-off**: multiplier R in the range 0 to 2^k-1 is randomly chosen and multiplied by propagation time, T_p , to get T_B

Pure ALOHA

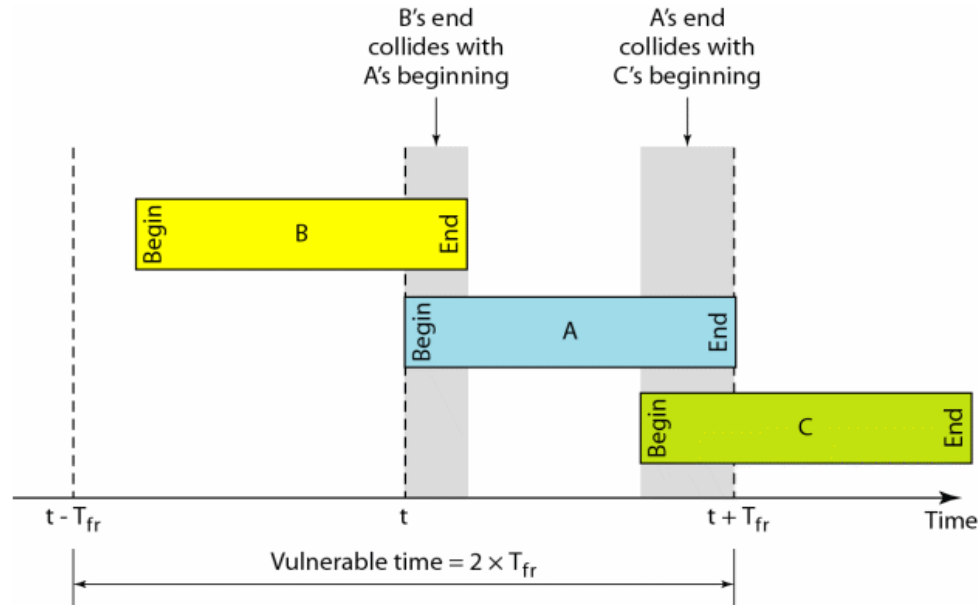
K : Number of attempts
 T_p : Maximum propagation time
 T_{fr} : Average transmission time for a frame
 T_B : Back-off time



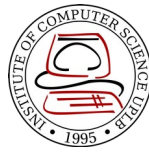
Pure ALOHA



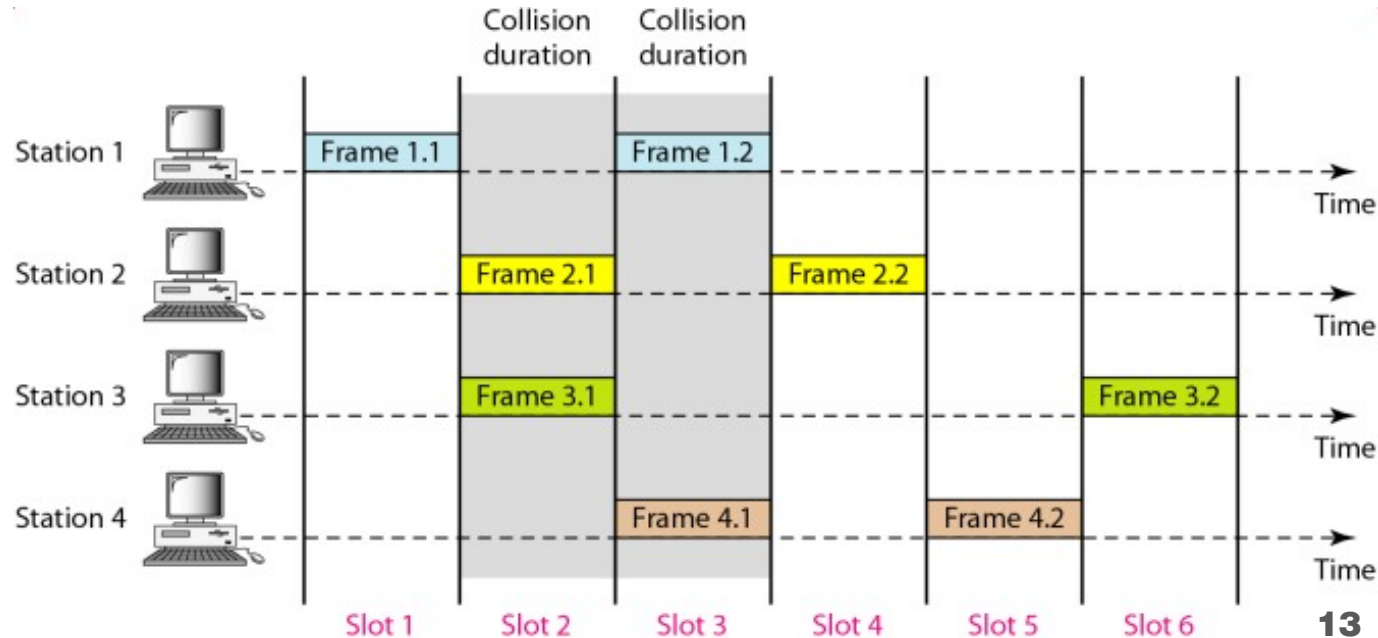
- Vulnerable time** – time that a possibility of a collision might occur; $2 \times T_{fr}$



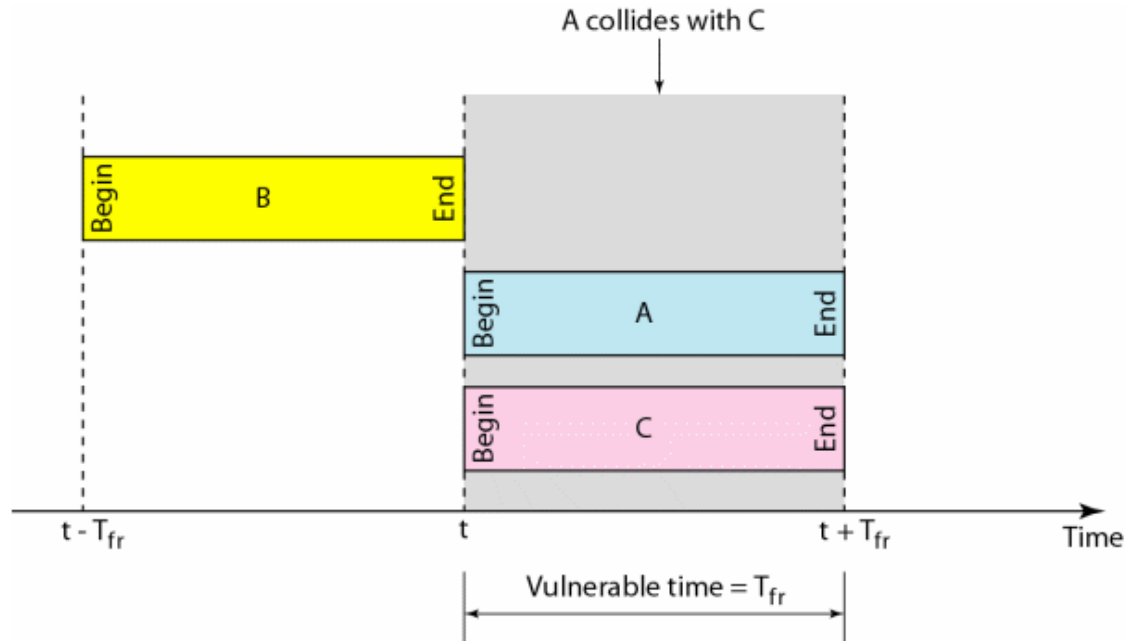
Slotted ALOHA



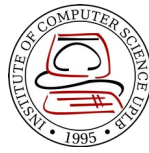
- Improves the efficiency of Pure ALOHA
- Divide the time into slots of T_{fr} and force the station to send only at the beginning of the time slot
- Vulnerable time is T_{fr}



Slotted ALOHA

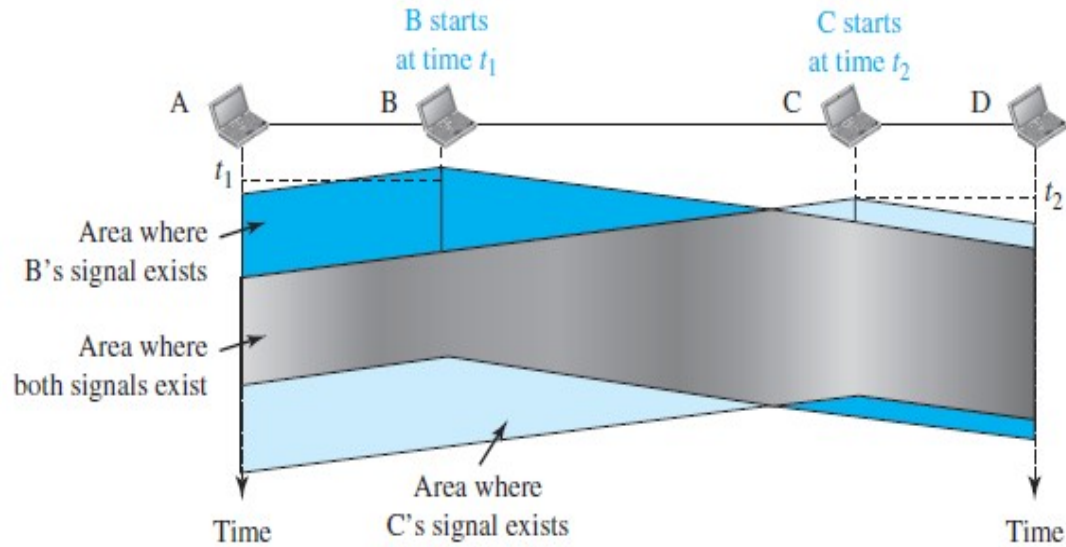


Carrier Sense Multiple Access (CSMA)

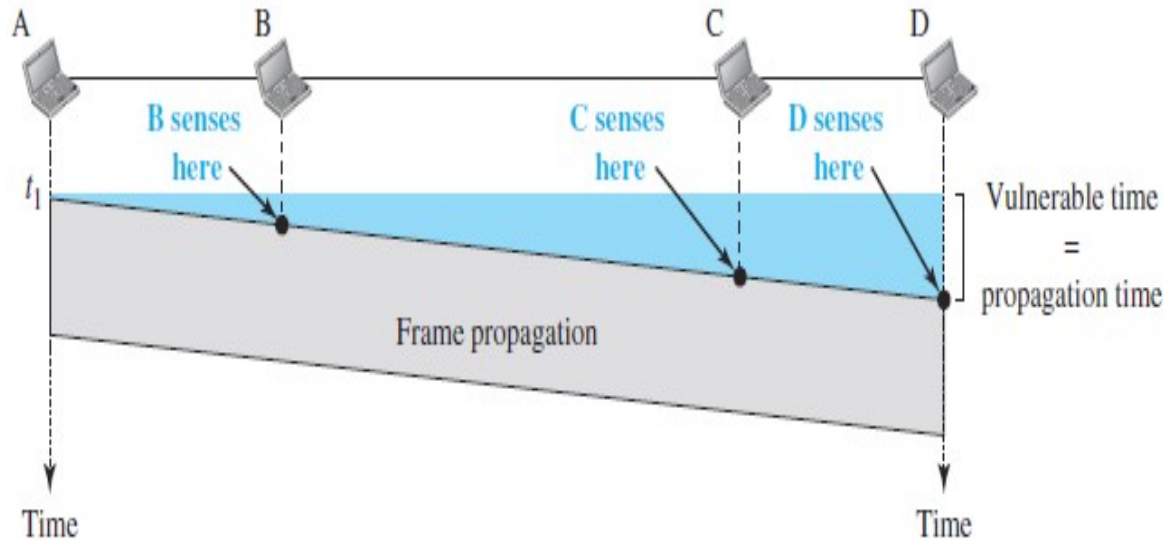


- The chance of collision can be reduced if a station senses the medium before trying to use it
- Each station first listens to the medium (check the state of the medium) before sending
- “sense before transmit”, “listen before talk”
- Can reduce possibility of collision but cannot eliminate it; because of propagation delay

Space/Time Collision



- Vulnerable time is the propagation time T_p

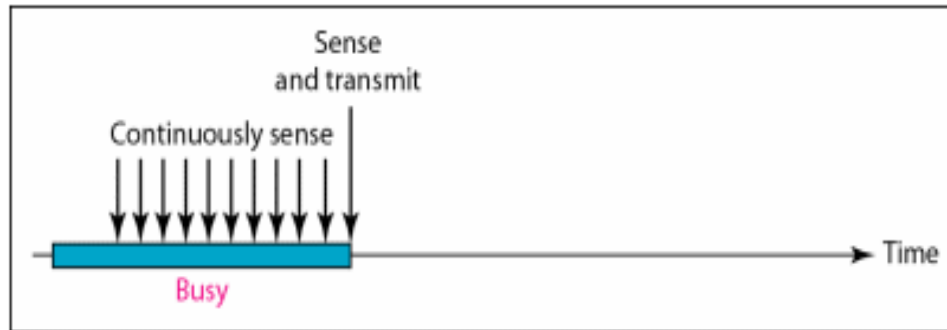


CSMA: 1-persistent

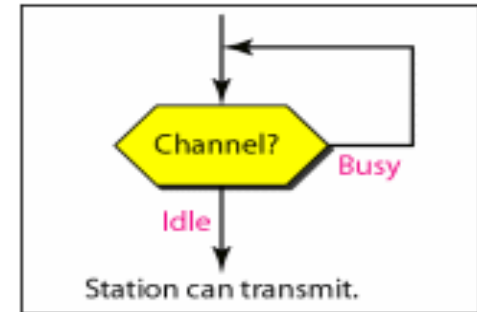


Station wishing to transmit listens to the medium

- If medium is idle, transmit immediately
- If medium is busy, continuously listen until medium becomes idle then transmit immediately



a. 1-persistent

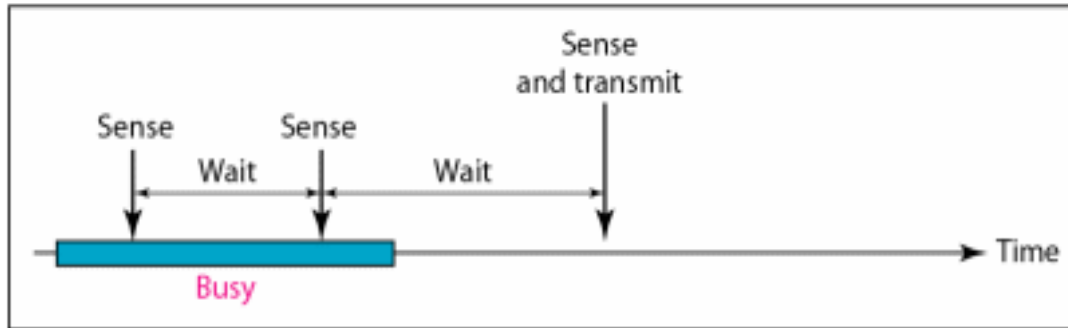


a. 1-persistent

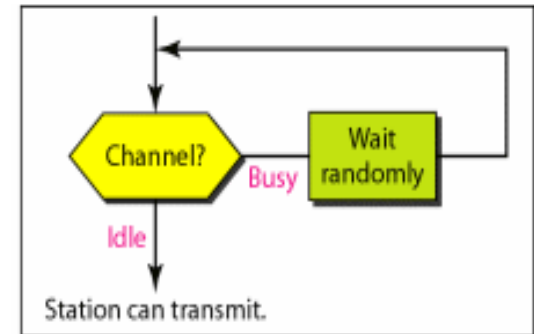
CSMA: Non-persistent



- Station wishing to transmit listens to the medium
 - If medium is idle, transmit
 - If medium busy, wait a random amount of time and repeat

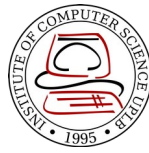


b. Nonpersistent

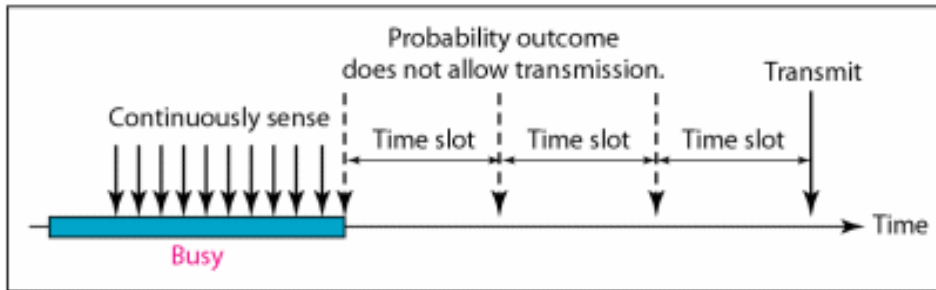


b. Nonpersistent

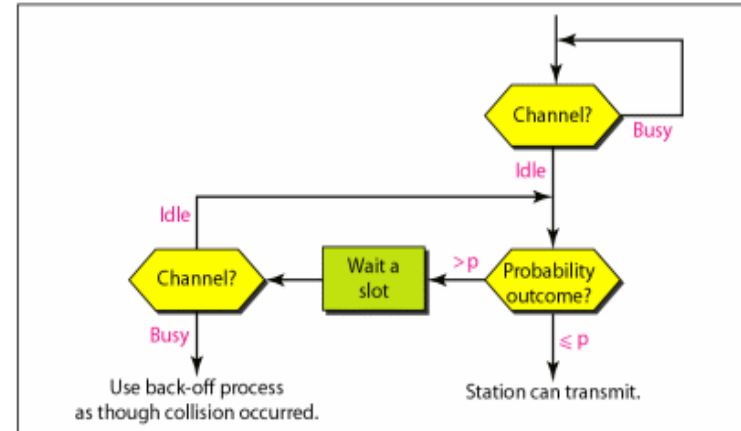
CSMA: P-persistent



- Time is divided to slots where each slot typically equals maximum propagation delay
- Station wishing to transmit listens to the medium
 - If medium is idle
 - transmit with probability (p), OR
 - wait one slot with probability ($1 - p$), then repeat
 - If medium busy, continuously listen until idle and repeat



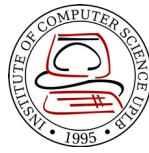
c. p-persistent



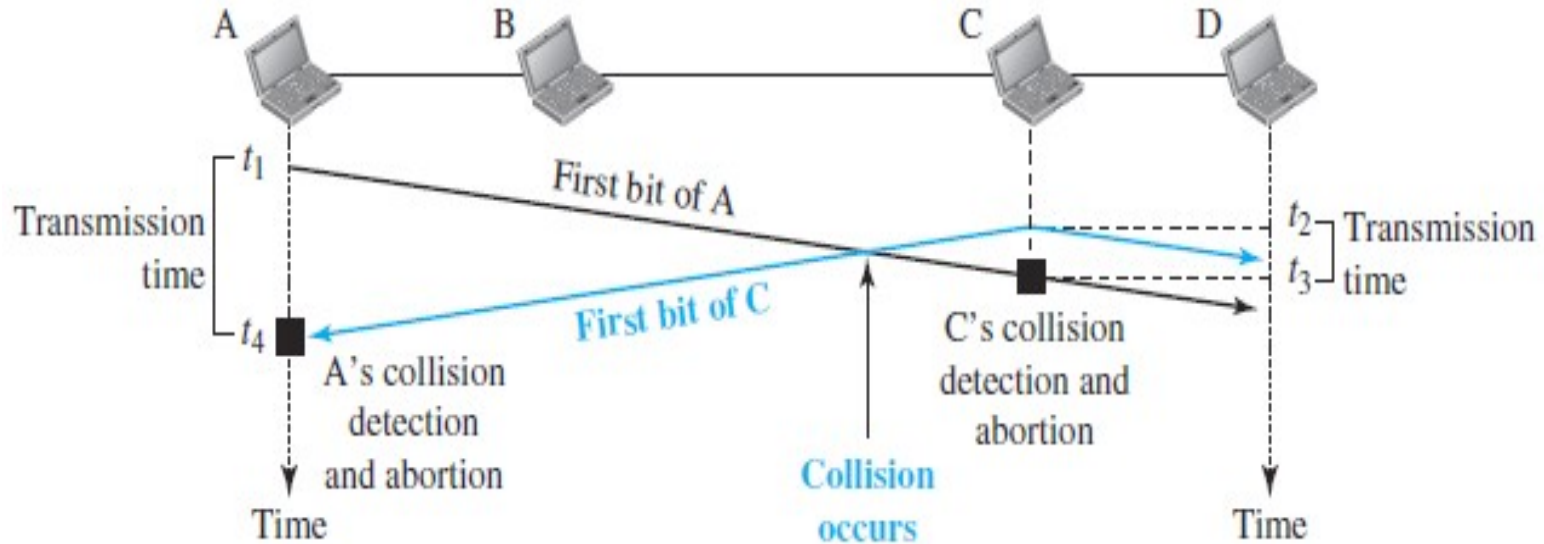
c. p-persistent

- Carrier Sense Multiple Access with Collision Detection
- A station monitors the medium after it sends a frame if the transmission was successful
- Widely used for bus topology LANs (IEEE 802.3 Ethernet)
- Minimum Frame Size: $T_{fr} = 2 \times T_p$
- Before sending the last bit of the frame, sending station must detect a collision

CSMA/CD



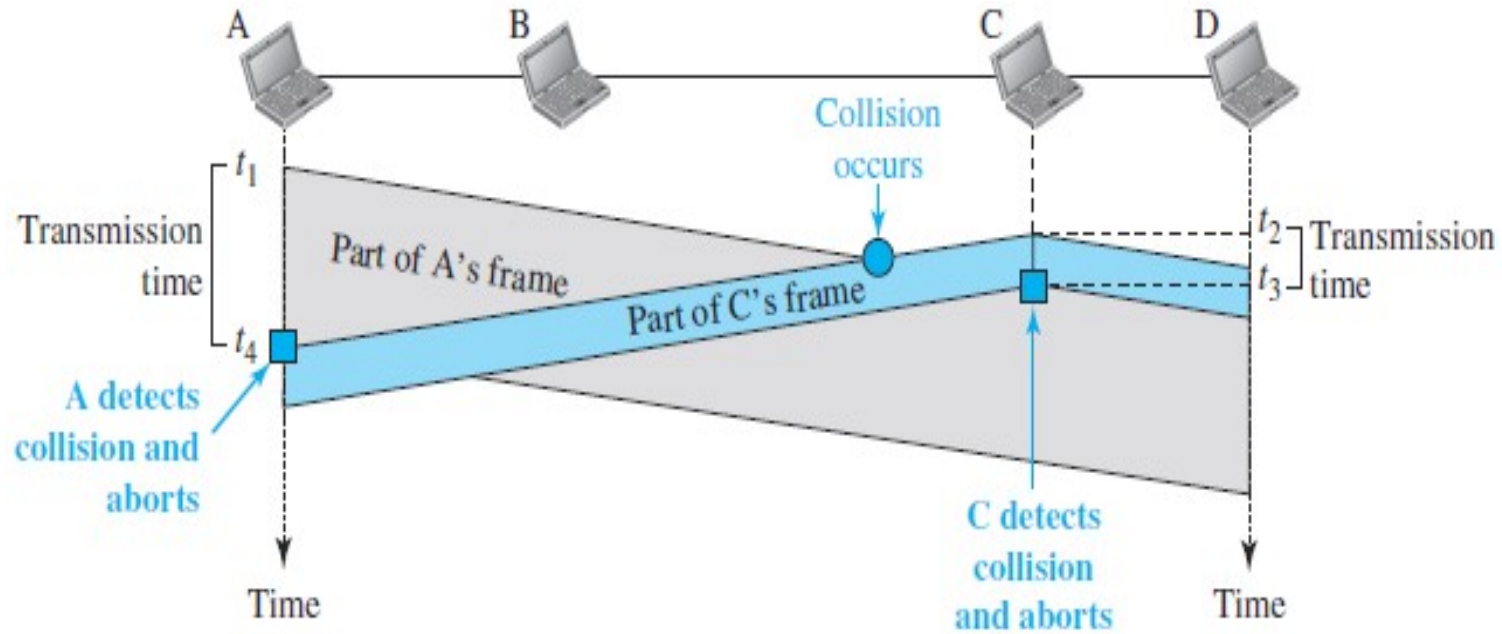
Collision of the first bits



CSMA/CD

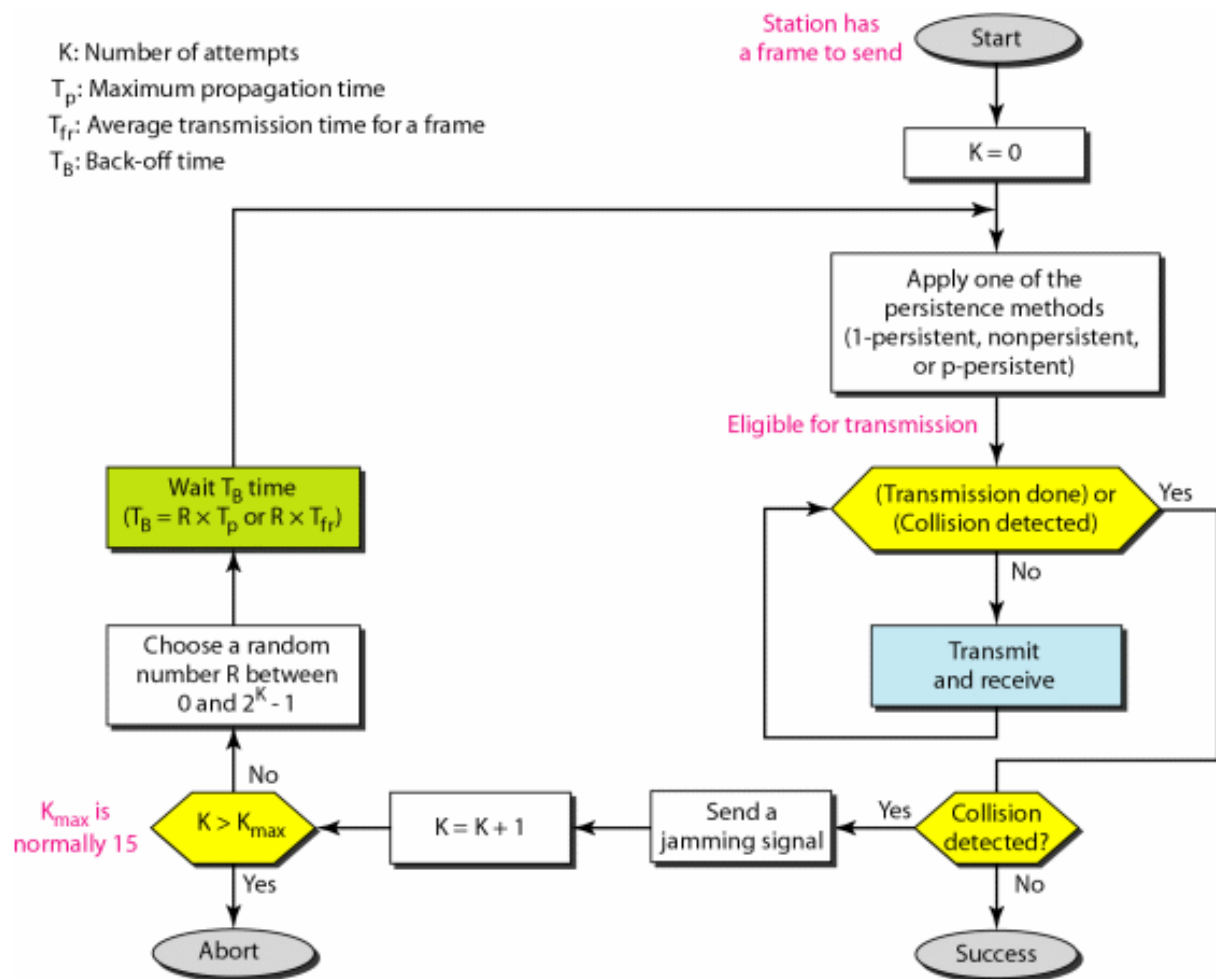


Collision and abortion

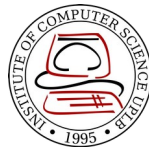


CSMA/CD

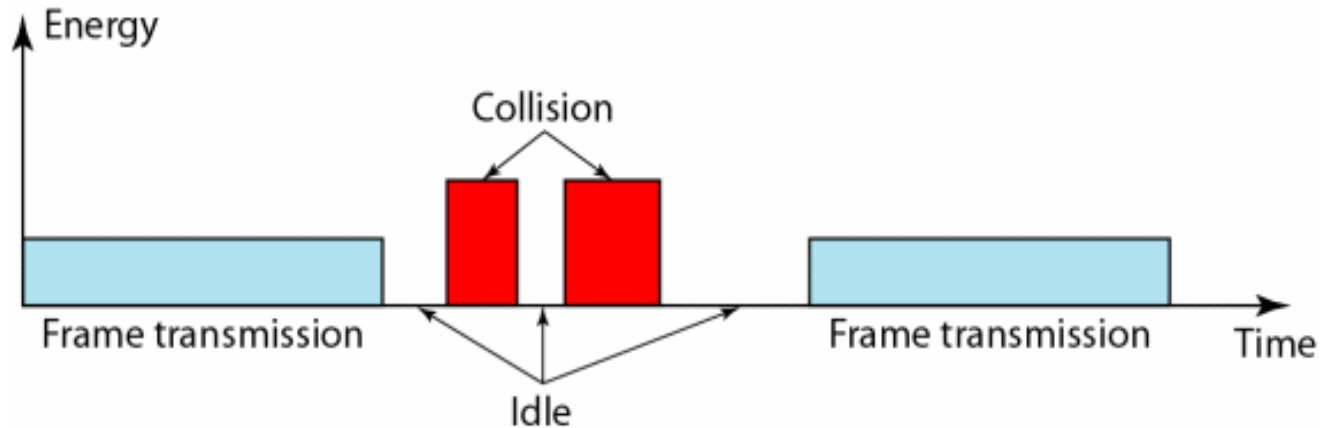
K : Number of attempts
 T_p : Maximum propagation time
 T_{fr} : Average transmission time for a frame
 T_B : Back-off time



CSMA/CD



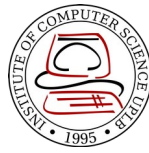
Energy level of the channel: **zero** (idle), **normal** (successful), **abnormal** (collision)



- Carrier Sense Multiple Access with Collision Avoidance
- A station needs to be able to receive while transmitting to detect collision
- **No collision**: station receives its own signal
- Collision: station receives its own signal and the signal transmitted by a second station
- In **wireless**, collisions must be avoided because it is difficult to detect
- Strategies: **Interframe space**, **contention window**, **ACKs**

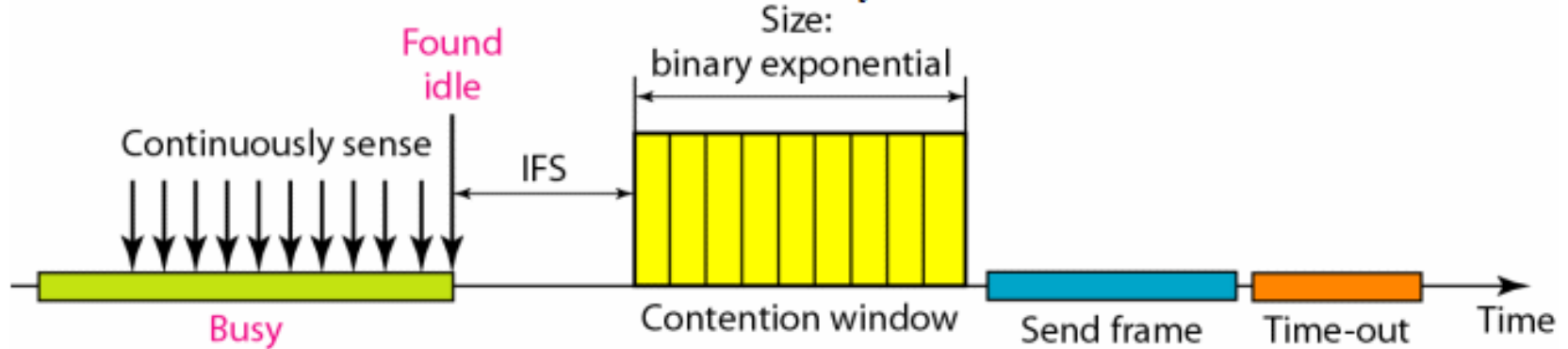
- **Interframe Space (IFS)**
 - Defer transmission even if channel is idle
 - Can be used to define the priority of a station
- **Contention Window**
 - An amount of time divided into slots
 - Station must sense the channel after each time slot
 - Does not restart the process if channel is busy
- **Acknowledgments**
 - guarantees receipt of frame

CSMA/CA

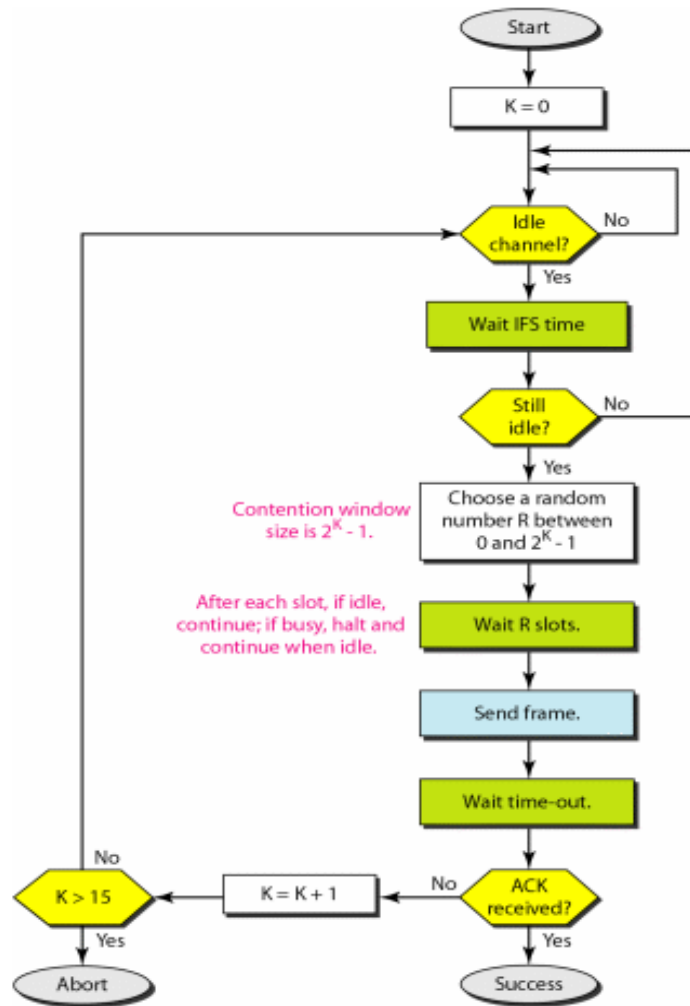


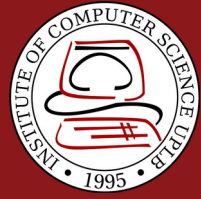
Timing

Avoid collisions on wireless network because they cannot be detected



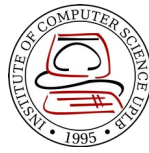
CSMA/CA





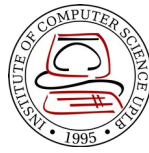
Controlled Access Protocols

Controlled Access

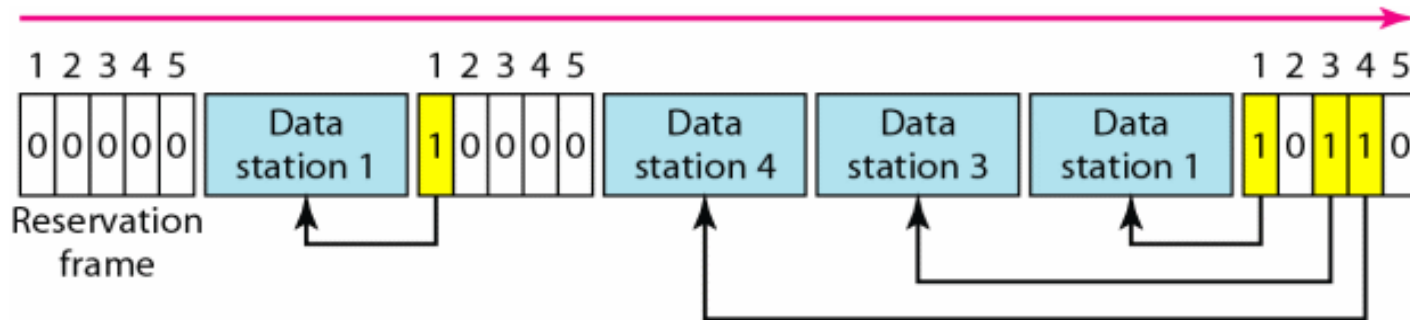


- Stations consult one another to find which station has the right to send
- A station cannot send unless it has been authorized by other stations
- Methods: **Reservation**, **Polling**, **Token Passing**

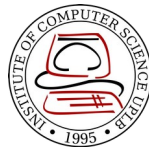
Reservation



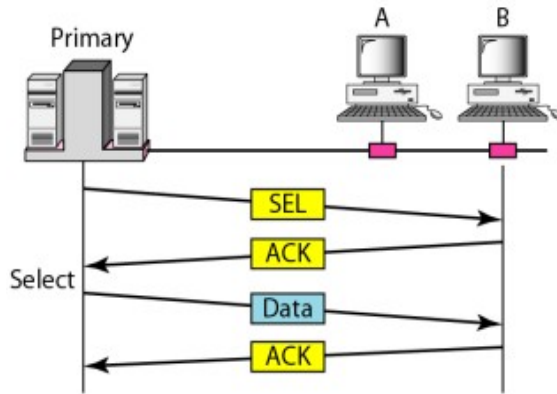
- Station must make a reservation before sending data
- Time is divided into intervals, at each interval, a reservation frame precedes the data frame
- N stations, N minislots in the reservation frame
- Station that have made reservations can send data frames after reservation frame



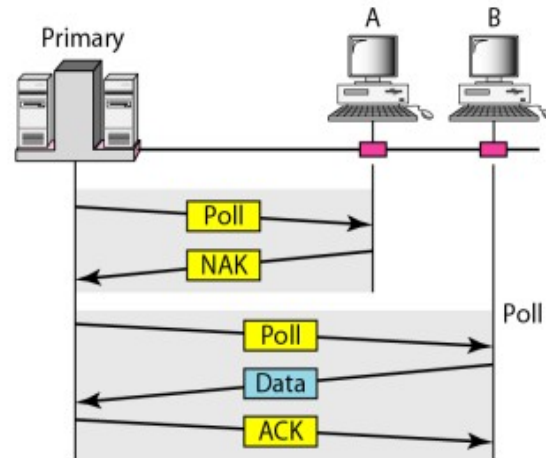
Polling



- Designated primary and secondary stations
- Data exchanges must be made through the primary device even when destination is a secondary
- Primary station controls the link

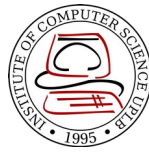


Select



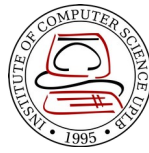
Poll

Polling



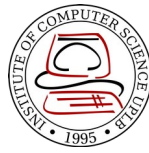
- **Select**
 - Used whenever primary has something to send
 - Must check if the receiver is ready receive
 - A SEL frame is send first, and station waits for ACK
- **Poll**
 - Used by primary to solicit transmissions from secondary devices
 - Must poll all stations

Token Passing



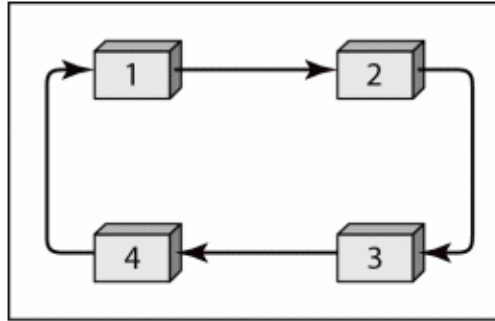
- Stations in a network is organized in a **logical ring**
- For each station, there is a **predecessor** and a **successor**
- Current station has access to channel, right to access has been passed from predecessor, will be passed to the successor
- A special frame, **token frame**, circulates around the ring
- Token frame gives right access to a station

Token Passing

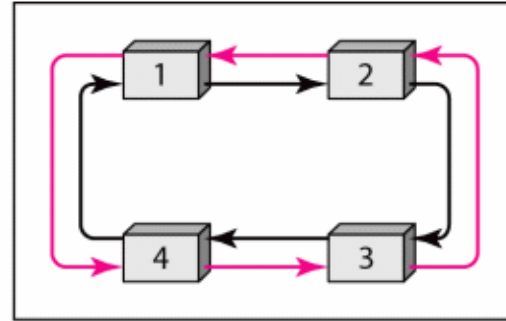


- Token Management
 - Stations must be limited in the time they can have possession of the token
 - Token must be monitored to ensure it has not been lost or destroyed
 - Assign priorities to stations
 - Low-priority stations must release the token to high priority stations

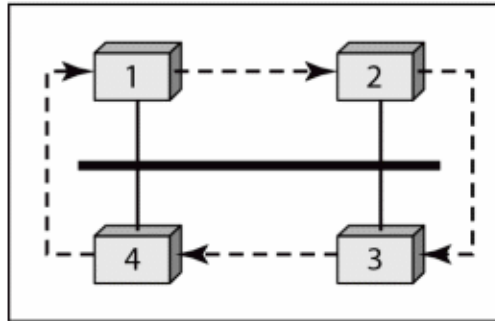
Token Passing



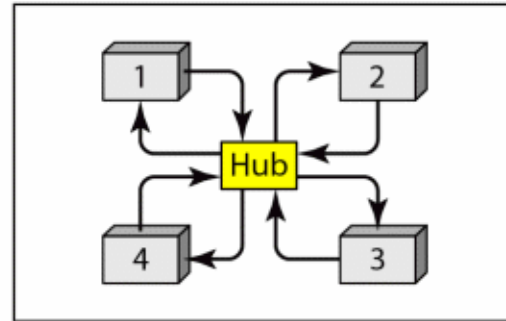
a. Physical ring



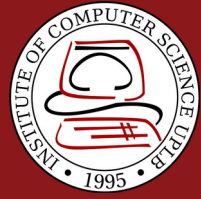
b. Dual ring



c. Bus ring



d. Star ring



Channelization Protocols

Channelization

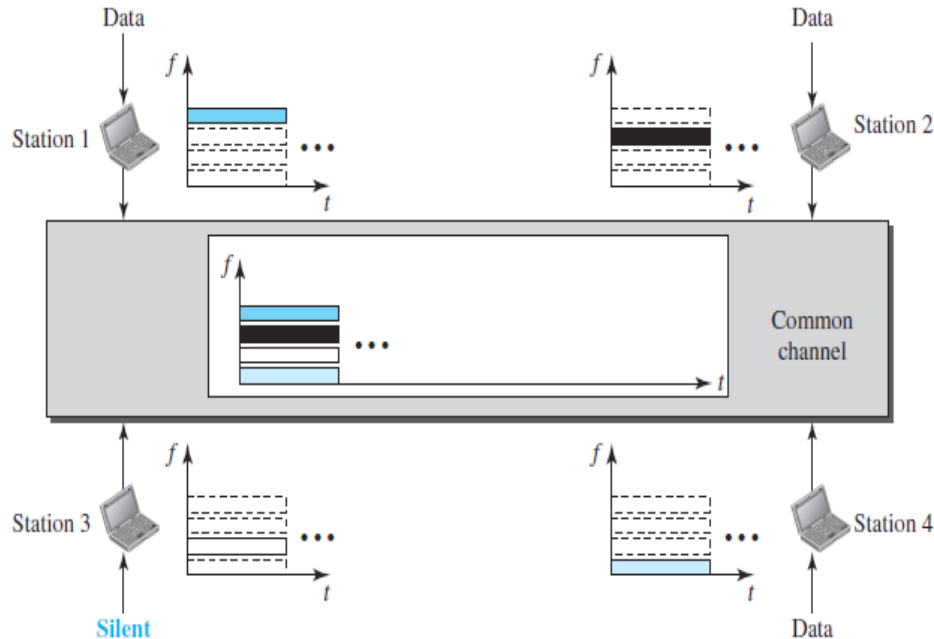


- Multiple-access method in which available bandwidth of a link is shared in time, frequency, or code between different stations
- Protocols: FDMA, TDMA, CDMA

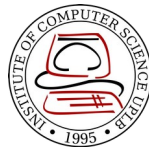
FDMA



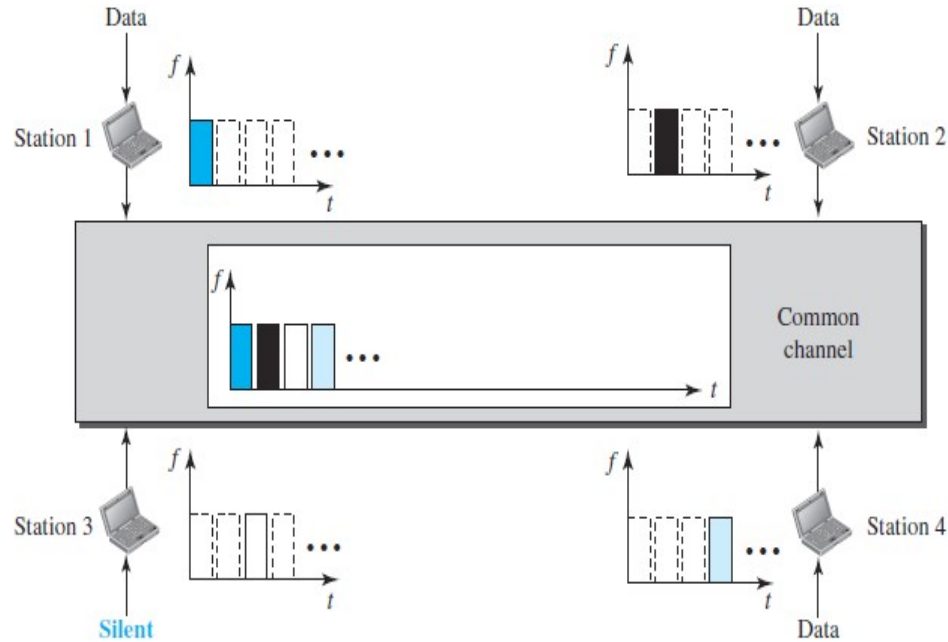
- Frequency-Division Multiple Access
- Available bandwidth is divided into bands separated by guard bands



TDMA

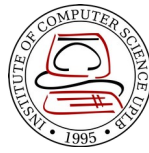


- Time-Division Multiple Access
- Share the bandwidth of the channel in time
- Sync problem present



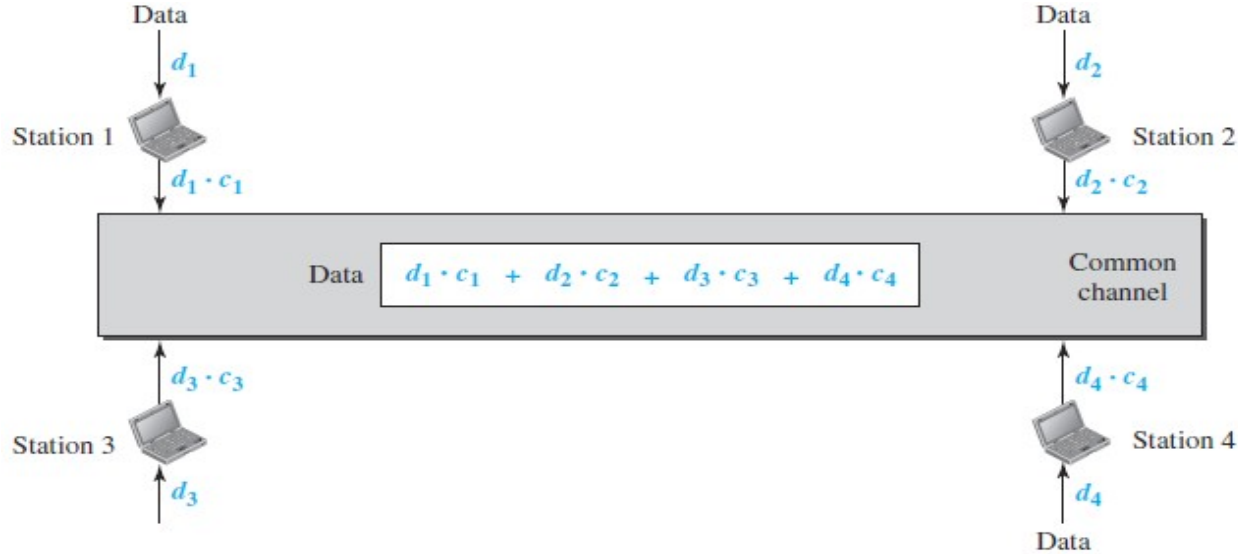
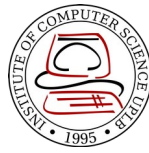
- Code-Division Multiple Access
- One channel carries all transmissions simultaneously
- Communication with different codes
- The stations receivers use these codes to recover the data for the desired station
- It differs from FDMA in that only one channel occupies the entire bandwidth; it differs from TDMA in that all stations can send data simultaneously
- Based on coding theory, each station is assigned a code, which is a sequence of numbers called chips

CDMA



- Assume we have four stations 1,2,3,4
- Data from station 1 are d_1 , from station 2 are d_2 , etc
- Code assigned to station 1 is c_1 , to the second is c_2
- Assume the code has the ff. properties
 - If we multiply each code by another we get 0
 - If we multiply each code by itself we get 4 (number of stations)

CDMA



$$\begin{aligned} \text{data} &= (d_1 \cdot c_1 + d_2 \cdot c_2 + d_3 \cdot c_3 + d_4 \cdot c_4) \cdot c_1 \\ &= d_1 \cdot c_1 \cdot c_1 + d_2 \cdot c_2 \cdot c_1 + d_3 \cdot c_3 \cdot c_1 + d_4 \cdot c_4 \cdot c_1 = 4 \times d_1 \end{aligned}$$

References



- Forouzan, B.A. 2013. Data Communications and Networking, 5th Ed. McGraw-Hill, New York.